MILITARY SPECIFICATION

MODULE ASSEMBLY, STORAGE, SHELF TYPE

This specification is approved for use within U.S. Marine Corps, Department of the Navy, and is available for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for manufacture and acceptance of the Module Assembly, Storage, Shelf Type. This item is the only one of its type.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

V-T-295	-	Thread, Nylon.
QQ-A-200/8	-	Aluminum Alloy 6061, Bar, Rod, Shapes, Tube and Wire, Extruded.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commandant of the Marine Corps, Headquarters Marine Corps (Code LMA-1), Washington, DC 20380 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC.7125

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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QQ-A-200/9	-	Aluminum Alloy 6063, Bar, Rod, Shapes, Tube and Wire, Extruded.
QQ-A-225/8	-	Aluminum Alloy 6061, Bar, Rod, Wire and Special Shapes, Rolled, Drawn or Cold Finished.
QQ-A-250/8	-	Aluminum Alloy 5052, Plate and Sheet.
QQ-S-781	-	Strapping, Steel and Seals.
PPP-B-601	-	Boxes, Wood, Cleated Plywood.
PPP-B-621	-	Box, Wood, Nailed and Lock-corner.

MILITARY

MIL-P-116	-	Preservation, Methods of.
MIL-B-117	-	Bag, Sleeve and Tubing - Interior Packaging.
MIL-T-704	-	Treatment and Painting of Material.
MIL-W-4088	-	Webbing, Textile, Woven Nylon.
MIL-S-5059	-	Steel, Corrosion-Resistant (18-8) Plate, Sheet and Strip (ASG).
MIL-C-5541	-	Chemical Conversion Coatings on Aluminum Alloys.
MIL-E-6060	-	Envelope, Packaging, Water Vapor Proof, Flexible.
MIL-S-7720	-	Steel, Corrosion, Resistant (18-8) Bars, Wire and Forging Stock.
MIL-G-12803	-	Gasket Material, Non-metalic.
MIL-P-19834	-	Plate, Identification, Metal Foil, Adhesive Back.
MIL-C-22750	-	Coating, Epoxy Polyamide.
MIL-W-45205	-	Welding, Gas Metal-Arc and Gas Tungsten-Arc, Aluminum Alloys, Readily Weldable for Structures, Excluding Armor.
MIL-P-53022	-	Primer, Epoxy Coating, Corrosion Inhibiting, Lead and Chromate Free.

STANDARDS

FEDERAL

FED-STD-595	-	Federal Standard Colors.

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-130	-	Identification Marking of U.S. Military Property.

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MIL-STD-2073-1A - DoD Material Procedures for Development and Application of Packaging Requirements.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

U.S. Marine Corps Logistics Base, Albany, GA
82A5048A1600 - Module Assembly, Storage, Shelf Type
82A5048A1131 - Shelf, Tie Down
82A5050A0100 - Rail Assembly and Installation, Mounting, Floor
82A5050A0110 - Peg Assembly, Modular Alignment
82A5050A0132 - Bar, Lock
82A5050A0138 - Shaft, Latch
82A5050A0140 - Head, Latch
82A5050A0142 - Handle, Latch
82A5050A0144 - Latch
82A5050A0150 - Rail Assembly and Installation, Mounting, Wall
82A5050A0160 - Post Assembly, Mounting

(Copies of specifications, standards, handbooks, drawings, publications, and other government documents required by suppliers in connection with specified procurement functions should be obtained from the contracting agency or as directed by the contracting officer).

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE

ANSI/AWS A2.4-79 - Welding and Nondestructive Testing, Symbols.
ANSI/AWS A3.0-85 - Welding Terms and Definitions.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

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AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM A108-81 - Steel Bars, Carbon, Cold Finished, Standard Quality

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for the associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3 and 6.4).

3.2 Materials. The materials and components shall be certified to conform to the applicable specification and standards as listed on drawing package 82A5048A1600 or required herein. The materials used may include reclaimed materials provided such materials will not jeopardize its intended use and performance. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner which will restore them to the same chemical composition and physical properties as the materials originally selected for use as specified herein.

3.3 Design. Module Assembly, Storage, Shelf Type shall be of the design specified on drawing package 82A5048A1600 (see 2.1.2).

3.4 Construction. Module Assembly, Storage, Shelf Type shall be of the construction and physical dimensions specified on drawing package 82A5048A1600 (see 2.1.2).

3.4.1 Frame assembly. The frame assembly shall be fabricated from Angle, Aluminum Alloy 6061-T6, QQ-A-200/8; Rod, Aluminum Alloy 6061-T6, QQ-A-225/8; Bar, Aluminum Alloy 6061-T6510, QQ-A-200/8; Aluminum Alloy 6032-T6, QQ-A-200/9; and shall be square in all directions within .010 inch per foot (see 4.4.2). All welds shall be in accordance with ANSI/AWS A3.0-85, A2.4-79 and MIL-W-45205, Class B (see 4.4.4). The frame assembly shall conform to specifications to such a degree of accuracy ensuring it will align to the peg and pin assemblies and lock using the swing latches as illustrated in the footprint drawing, Figure 1 (see 4.4.3).

3.4.2 Shelf assembly. The shelf shall be in accordance with drawing 82A5048A1131. Attached to the shelf shall be two bar locks.

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The support, shelf shall be fabricated from Angle, Aluminum Alloy 6063-T6, QQ-A-200/9. The slide drawer extension (3.5) Structural Members shall be made from Aluminum Alloy with a minimum yield strength of 35,000 psi, slide ball raceways shall have a minimum ultimate tensile strength of 200,000 psi. The slide drawer extension (3.5) Component Parts shall be of 300 Series Stainless Steel per MIL-S-5059 or MIL-S-7720 Carbon Steel per ASTM A108-81 and or Aluminum Alloy. The slide drawer extension shall have a load rating (per pair) at 22 inch extension of 200 lbs. The slide supports shall be fabricated from Sheet, Aluminum Alloy, 5052-H32, QQ-A-250/8. Attached to each slide support shall be a latch. Each slide support shall be securely bolted to the frame in two places. Each slide drawer extension (3.5) shall be securely screwed to the slide support in eight places. Each shelf support shall be securely screwed to the slide drawer extension (3.5) in six places. The shelf shall be securely bolted to the shelf supports in 22 places. Installed the shelf assembly shall operate smoothly and freely with no binding (see 4.4.5).

3.4.3 Shelf assembly, stationary. The shelf assembly, tie down, shall be in accordance with drawing 82A5048A1131. The shelf shall be securely screwed to the frame assembly in 16 places (see 4.4.6).

3.4.4 Strap assembly, tie down. The strap assembly, tie down shall be fabricated from Strap, Webbing, MIL-W-4088, Type II, Class I, 2500 lbs minimum strength, color Gray; Strap, Webbing MIL-W-4088, Type XXVII, Class I, 6500 lbs minimum strength, color Gray, sewn with Thread per Fed Spec V-T-295, Type II, Class A, Type B, color Black per FED-STD-595. Attaching hardware will consist of Rotating End Fitting; Plate, Threading; Hook, Flat; and Fastener, Flat Hook. Attaching hardware shall be cold Rolled Steel, 300 lbs minimum strength with HD Zinc Plate Bronze Chromate Finish. The strap assembly shall be fabricated to have a minimum strength of 300 lbs (see 4.4.7).

3.4.5 Bracket mounting. The bracket mounting, shall be fabricated of Angle, Aluminum Alloy, 6061-T6, QQ-A-200/8 and shall be securely bolted to the frame in four places to such a degree of accuracy ensuring it will align to the footprint as described in Figure 1 (see 4.4.8).

3.4.6 Decal. The decal shall be fabricated from .003 inch thick aluminum alloy foil, photographically sensitized, adhesive back in accordance with MIL-P-19834, Type I. Requirements and methods of marking the decal shall be in accordance with MIL-STD-130. Unless otherwise specified all characters shall be .093 inch high on finished decal, color shall be in accordance with MIL-P-19834, Style III (see 4.4.9).

3.5 Finish.

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3.5.1 Cleaning. Prior to painting metal surfaces shall be cleaned and treated per MIL-T-704, Type G, and MIL-C-5541, Class 3.

3.5.2 Priming. After metal surfaces have been cleaned as required they shall be primed per MIL-P-53022, with a minimum dry film thickness of .001 inch in one coat.

3.5.3 Painting. Final finish painting shall have a minimum dry coat film thickness of .0018 inch in two coats per MIL-C-22750, Type II, Color No. 24533 per FED-STD-595 (see 4.4.12).

3.6 Interchangeability. All parts having the same part number shall be functionally and dimensionally interchangeable. Interchangeable parts are defined as two or more like parts possessing such functional and physical characteristics as to be equivalent in performance and durability, and capable of being exchanged one for the other without alteration of the parts themselves or adjoinin parts and without selection for fit or performance.

3.7 Workmanship. The Module Assembly, Storage, Shelf Type shall meet all design, fabrication, assembly and finish requirements of this specification. Particular attention shall be given to: removal of burrs and sharp edges; accuracy of dimensions; thoroughness of welding; painting; alignment; tightness of screws, bolts, and rivets. Failure to ensure these measures have been taken will be considered adequate basis for rejection of items being of inferior quality for the purpose intended.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1. Responsibility for compliance. All items must meet all requirements of section 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspectior requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

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4.1.2 Test equipment and inspection facilities. When specified (see 6.2) the contractor shall insure that test and inspection facilities of sufficient accuracy, quality and quantity are established and maintained to permit performance of required inspections.

4.2 Classification of inspections. The inspections specified herein are classified as follows:

a. First article inspection. First article inspection consists of examinations and tests performed on samples which are representative of the production item after award of a contract to determine that the production item conforms to the requirements of this specification (see 4.3).

b. Quality conformance inspection. Quality conformance inspection consists of examinations and tests performed on individual products to determine conformance of the products with the requirements set forth in this specification (see 4.4).

c. Visual inspection. Visual inspection consists of examinations to determine conformance with the requirements set forth in this specification (see 4.5).

4.3 First article inspection. The first article inspection of the Module Assembly, Storage, Shelf Type shall consist of examination and tests for all the requirements of this specification.

4.3.1 First article sample. Unless otherwise specified as soon as practical after the award of the contract, the contractor shall submit quantity(s) specified in the contract for Module Assembly, Storage, Shelf Type. The modules shall be representative of the construction, workmanship, components, and materials to be used during production. Approval of the first article inspection samples or the waiving of the first article inspection does not preclude the requirements for performing the quality conformance inspection. The first article inspection samples shall be furnished to the Government as directed by the Contracting Officer (see 6.2).

4.3.2 Sample disposition. Upon completion of the first article inspection, sample(s) will be disposed of in accordance with Government shipping instructions.

4.3.3 First article inspection routine. First article inspection shall consist of those items listed in Table I.

4.4 Quality conformance inspection. Quality conformance inspections shall be conducted in accordance with MIL-STD-105 and consist of those items listed in Table I.

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TABLE I. Test and Inspection

Requirement	Requirement Paragraph	Test Paragraph
Condition and certification of materials	3.4.1, 3.4.2 3.4.4, 3.4.6	4.4.1
Dimensional check of frame assembly	3.4.1	4.4.2
Verification Frame Assembly fits footprint	3.4.1	4.4.3
Welding	3.4.1	4.4.4
Shelf assembly	3.4.2	4.4.5
Shelf assembly, stationary	3.4.3	4.4.6
Strap assembly, tie down	3.4.4	4.4.7
Bracket, mounting	3.4.5	4.4.8
Decal	3.4.6	4.4.9
Cleaning of aluminum	3.5.1	4.4.10
Priming	3.5.2	4.4.11
Painting	3.5.3	4.4.12
Packaging	5.2	4.4.13
Marking	5.3	4.4.14
Hardware	3.4.2, 3.4.3 3.4.5	4.4.15

4.4.1 Condition and certification of material. Material used in construction of the Module Assembly, Storage, Shelf Type (sheet, roll, and bar aluminum) shall be certified to be of the specified material that has no etching, scaling or pitting. Fabricated pieces shall not have sharp edges or burrs. Material shall not be cracked, malformed or split.

4.4.2 Frame dimensions. The frame assembly shall be measured and shall be square in all directions to within .010 inch per foot.

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4.4.3 Footprint fit. The frame assembly shall be inspected for alignment to the peg and pin assemblies and lock using the swing latches as illustrated in the footprint drawing, Figure 1.

4.4.4 Welding. Welding areas shall be inspected for cleanliness and be free of dirt, sand, scale and oil prior to welding. All welds shall be inspected for compliance with ANSI/AWS A3.0-85, A2.4-79, and MIL-W-45205 Class B.

4.4.5 Shelf assembly. The shelf assembly shall be inspected for smooth and free operation with no binding.

4.4.6 Shelf assembly, stationary. The shelf assembly shall be inspected for secure attachment to the frame assembly in 16 places.

4.4.7 Strap assembly, tie down. The strap assembly tie down shall tested for a minimum strength of 300 lbs.

4.4.8 Bracket mounting. The bracket mounting shall be inspected for secure attachment to the frame to such a degree of accuracy to align with the footprint as described in Figure 1.

4.4.9 Decal. The decal shall be inspected for attachment to the Module Assembly, Storage, Shelf Type, as specified on the drawings and color to be in accordance with MIL-P-19834, Style III.

4.4.10 Cleaning of aluminum. Cleaning of aluminum surfaces prior to painting shall be inspected for conformance with MIL-T-704, Type G, and MIL-C-5541, Class 3.

4.4.11 Priming. Priming shall be inspected for conformance with MIL-P-53022.

4.4.12 Painting. Painting of the Module Assembly, Storage, Shelf Type, shall be inspected for conformance with MIL-C-22750, Type II, Color No. 24533 per FED-STD-595 with final finish coat having a minimum dry coat thickness of .0018 inch in two coats. Coatings shall be inspected to have a level adherent continuous and uniform film without runs, wrinkles, streaks, or areas of no film.

4.4.13 Packaging. Packaging shall be inspected for conformance with MIL-STD-2073-1A.

4.4.14 Marking. Marking on the shipping container shall be inspected for conformance with MIL-STD-129.

4.4.15 Hardware. All hardware shall be inspected to be in place and securely fastened.

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4.5 Visual inspection. The Module Assembly, Storage, Shelf Type shall be examined for defects listed in Table II. The defects shall be classified as specified in Table II.

TABLE II. Classification of Defects for Visual Inspection

Defect	Classification	
	Major	Minor
Frame or components misaligned	X	
Welds cracked or incomplete	X	
Metal components dented, misshaped, distorted or otherwise defective	X	
Sharp burrs or metal slivers	X	
Hardware missing	X 1/	
Hardware not securely fastened		X
Decal mounted incorrectly		X

1/ Major when it affects the strength of attachment; otherwise, it is to be classified as a minor defect.

5. PACKAGING

5.1 Preservation. Preservation shall be level A in accordance with MIL-STD-2073-1A.

5.1.1 Unit packs. Each module shall be individually unit packed in accordance with submethod 1A-16 (Floating Bag Sealed), of MIL-P-116. Modules preserved, wrapped, and cushioned as required in paragraph 3.6.3 of MIL-P-116, shall be enclosed in a sealed bag conforming to MIL-B-117, type I or II; class E, F, or G, Style 1 or 2. Bags in accordance with MIL-E-6060 shall be used for sizes larger than can be made in accordance with MIL-B-117. The bag shall be provided with gasketed holes to permit the installation of the barrier over and around the fasteners used to secure the item(s) to interior supports or to one face of the container. Gaskets shall be attached to the barrier with adhesive. Unless otherwise specified, material for gaskets shall conform to MIL-G-12803 (see paragraph 6.2.1 (g)). Material for gaskets shall not contain asbestos.

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5.2 Packing. Packing shall be level B in accordance with MIL-STD-2073-1A.

5.2.1 Level B. Modules, preserved as specified in 5.1, shall be packed in wood containers conforming to PPP-B-601, overseas type of PPP-B-621, class 2. Closure and strapping shall be in accordance with the applicable container specification except that metal strapping shall conform to QQ-S-781, type I, finish A.

5.3 Marking. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The Module Assembly, Storage, Shelf Type and other modules will be utilized in furnishing an Electronic Maintenance Complex (EMC) which will support a given mission. The number and type modules installed in any EMC or group of EMCs, at any given time, will depend directly on the user's immediate needs. This module provides for general storage of equipment.

6.2 Ordering data.

6.2.1 Acquisition requirements. Procurement documents shall specify the following:

a. Title, number, and date of this specification.

b. Quantity desired.

c. When first article testing is required (see 3.1).

d. When test equipment and inspection facilities are required (see 4.1.2).

e. When quality conformance testing is required (see 4.4).

f. Name and address of the first article inspection facility.

g. Level of preservation, packaging, and packing required (see section 5).

h. Whether any special markings are required.

6.3 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL) the data requirements identified below shall

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be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DOD FAR Supplement, Part 27, Sub-Part 27.410-6 are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

Data Requirements	Applicable DID Number	Options
First Article Inspection Procedure	DI-T-4901	-------
First Article Inspection Report	DI-T-4902	-------
Production/Acceptance Inspection Procedures	DI-T-4903	-------
Production Inspection Report	DI-T-4904	-------

(Copies of DID's required by contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

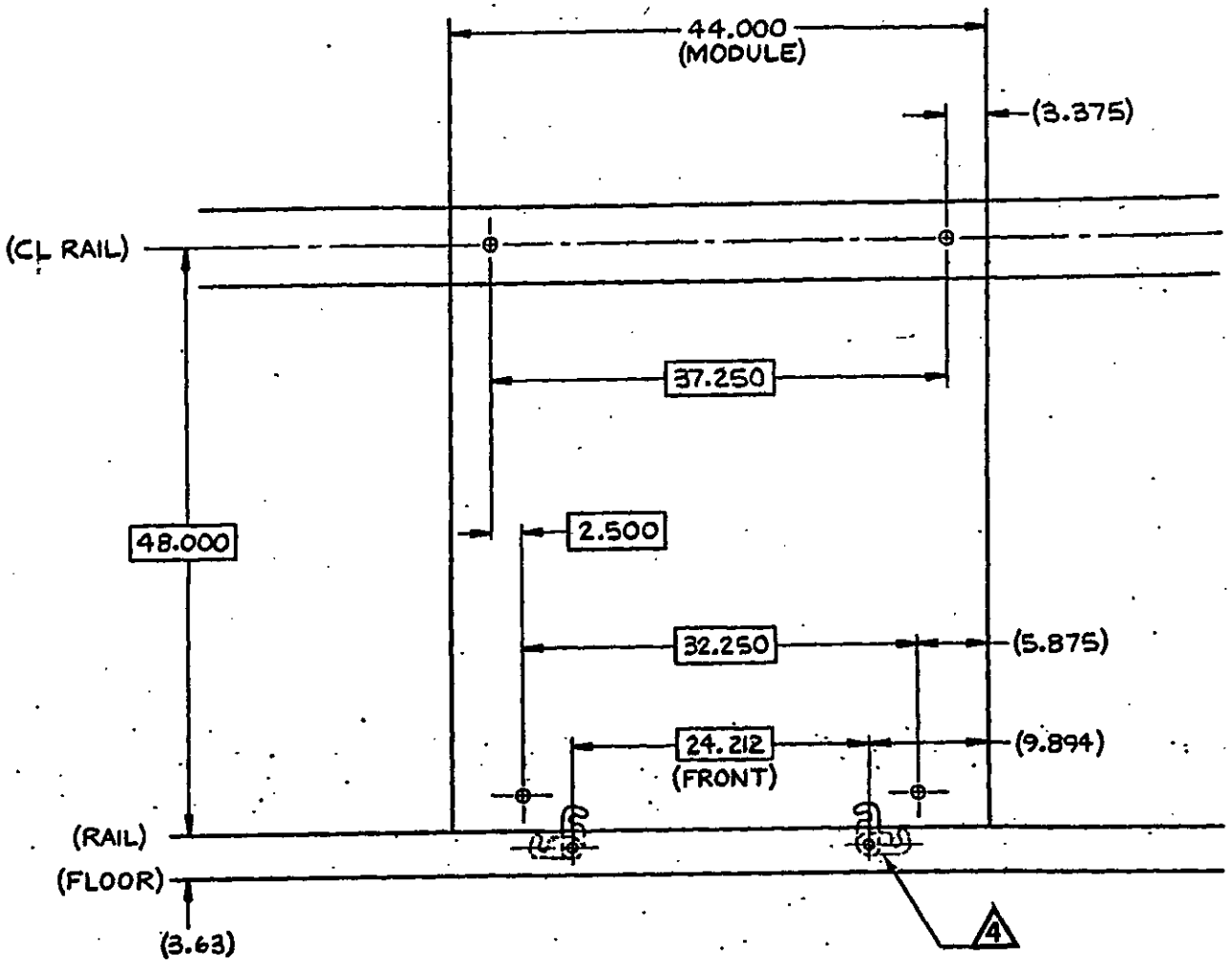
6.4 First article. The first article is to be examined and tested for approval at the contractor's plant or at an independent commercial testing laboratory acceptable to the procuring activity. First article tests shall be witnessed by a Government Representative of the procuring activity. Approval of first articles shall be by the procurement contracting officer.

6.5 Subject term (key word) listing. The following subject term (key word) listing is furnished for retrieval searches.

Assembly, frame
Assembly, module, storage, shelf type
Assembly, shelf
Assembly, shelf, stationary
Assembly, strap, tie down

Preparing activity:
Navy-MC
Project No. 7125-N124

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FRONT VIEW

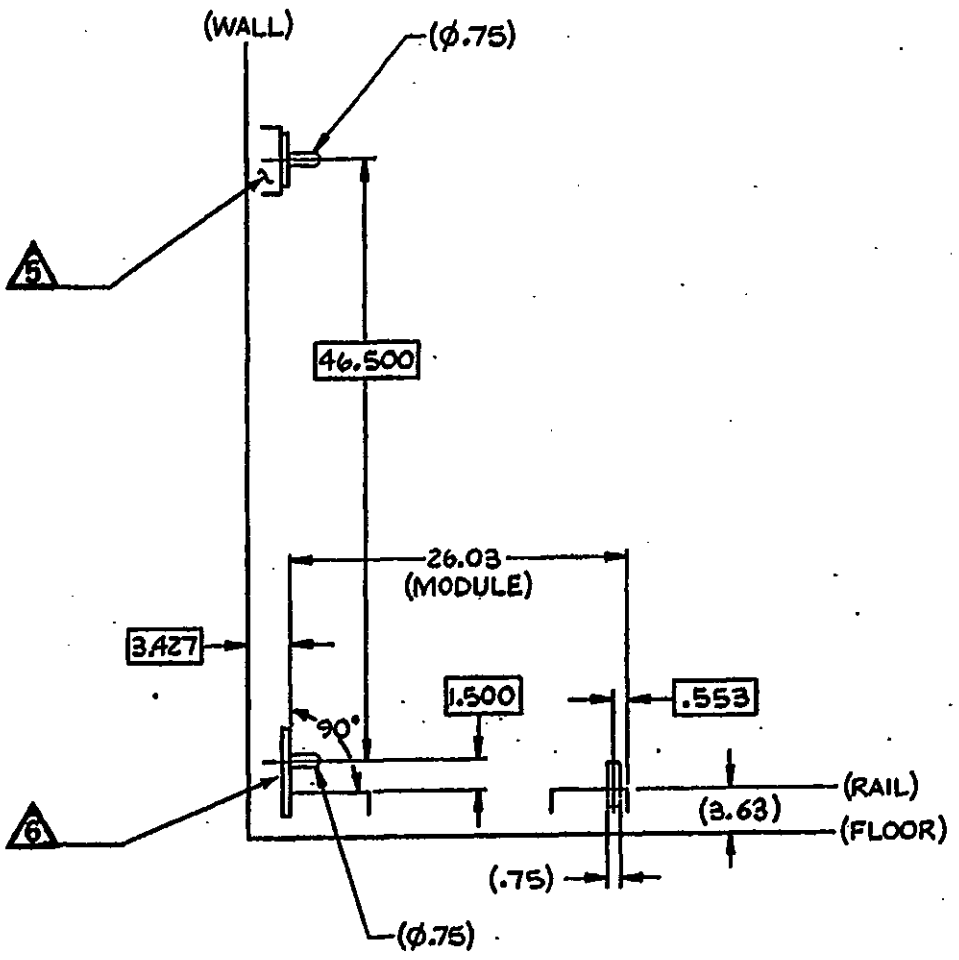
NOTES:

1. Dimensions are in inches.
2. Unless otherwise specified, tolerance is .XX ±.03, .XXX ±.010.
3. Perpendicularity equals ±1°.

4 REF DWGS: 82A5050A0100 (RAIL ASSEMBLY AND INSTALLATION, MOUNTING, FLOOR)
82A5050A0132 (BAR, LOCK)
82A5050A0138 (SHAFT, LATCH)
82A5050A0140 (HEAD, LATCH)
82A5050A0142 (HANDLE, LATCH)
82A5050A0144 (LATCH)

Figure 1. (Page 1 of 2)
Module Assembly Footprint

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SIDE VIEW

5 REF DWGS: 82A5050A0150 (RAIL ASSEMBLY AND INSTALLATION, MOUNTING, WALL)
82A5050A0160 (POST ASSEMBLY, MOUNTING)

6 REF DWGS: 82A5050A0100 (RAIL ASSEMBLY AND INSTALLATION, MOUNTING, FLOOR)
82A5050A0110 (PEG ASSEMBLY, MODULAR ALIGNMENT)

Figure 1. (Page 2 of 2)
Module Assembly Footprint

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